

FILL-IN-THE-BLANK (10)

Each blank is worth 1 point.

1. A camera captures uncompressed RGB images at a resolution of 1000×1000 pixels, with 8 bits per channel.
 - (a) How many **megabytes (MB)** are required to store one raw frame? Recall that $1\text{MB} = 1,000,000$ bytes, $1\text{byte} = 8$ bits. _____ MB are required.
 - (b) To reduce storage, you may reduce either the spatial resolution or the intensity resolution (bits per channel). If you halve the width and height, what should the new bit-depth per channel be, to achieve the same reduction in storage? _____ bits per channel.
2. A drone captures an image of a dark forest at night. 94.1% of the pixels have intensity values uniformly distributed between 10 and 15. The remaining 5.9% of pixels represent bright lights and stars in the sky with an intensity of 255.
 - (a) You opt to apply histogram equalization to this image. How many unique intensity values will the image have afterwards? _____ unique intensity values.
 - (b) Assume fractional values of $240/255$ and $15/255$ for the proportion of dark and light pixels, respectively. What would pixels with an original intensity value of 12 map to? 12 maps to _____
3. Consider the following 5×5 greyscale image of a diagonal edge (1=white, 0=black).

$$I = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Let the notation (r,c) indicate (row, column), with indexing starting at 1. Use Cartesian coordinates (+x points right, +y points upwards) and that angles are measured counterclockwise from the positive x-axis.

Recall the two Sobel kernels

$$S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad S_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

- (a) You compute image gradients without any border padding. How many unique gradient vectors are present in this image? _____ unique gradient vectors.
- (b) What is the gradient magnitude at element (4,4) of the image matrix? Indicate square roots as $(x)^{0.5}$. The gradient magnitude is _____.
- (c) What is the gradient orientation at element (3,3)? Indicate in degrees. The orientation is _____ degrees.
4. You are building a texture classification system using a texton dictionary and brute-force nearest neighbour-classification based on a sum-of-squared differences.

Your training dataset contains **100 images**, each of spatial size **100 × 100** pixels. A filter bank with **10 filters of size 11x11** is applied to every image. Assume that there is no border padding. The pixel-wise filter responses are clustered using K-means to learn a dictionary of **50 textons**. Each image is represented by a single histogram with one bin per texton.

Fill in the following blanks. If the answer cannot be determined from the information given, put -1.

- (a) The dimensionality of feature vectors fed into the K-means clustering to learn the dictionary is _____.
- (b) The total number of feature vectors fed into the K-means clustering to learn the dictionary is _____.
- (c) To classify each image, the number of squared-difference terms which must be computed is _____.

MULTIPLE CHOICE (10)

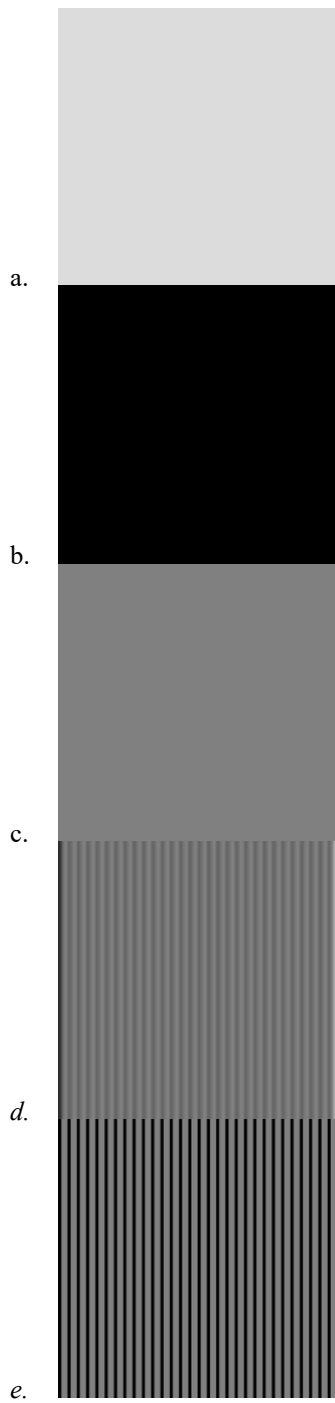
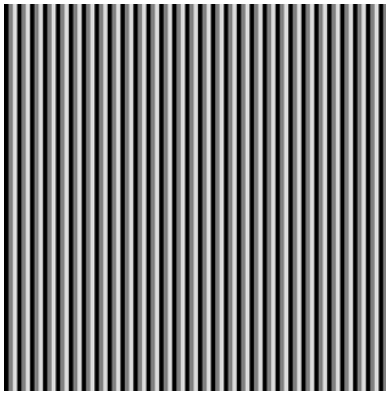
Each MCQ is worth 1 point.

1. You are building an algorithm to detect a specific orange basketball in a video feed. The lighting conditions vary significantly (e.g., bright sunlight vs. shadows), which changes the pixel intensity. Which colour representation would be most robust for defining the "orange" colour independent of brightness?
 - a. Raw RGB values
 - b. Greyscale intensity, based on RGB weights
 $\{w_R = 0.8, w_G = 0.2, w_B = 0\}$
 - c. Greyscale intensity, based on RGB weights
 $\{w_R = 0.299, w_G = 0.587, w_B = 0.114\}$
 - d. Normalized RGB
 - e. The 'V' channel in HSV

2. Recall the image formation model $f(x, y) = i(x, y) \cdot r(x, y)$. A flat, uniform piece of white paper lies on a desk and is illuminated by a desk lamp positioned to the left. As you scan the pixels from the left side of the paper to the right side, how do the model components change?
 - a. $r(x, y)$ decreases; $i(x, y)$ remains constant, so
 - b. $r(x, y)$ remains constant; $i(x, y)$ decreases
 - c. $r(x, y)$ increases; $i(x, y)$ decreases
 - d. $r(x, y)$ decreases; $i(x, y)$ increases
 - e. $r(x, y)$ remains constant; $i(x, y)$ remains constant

3. An image has a minimum intensity of 100 and maximum of 200. After applying a point processing operation, some resulting pixel values become negative. Which operation was definitively applied?
 - a. Histogram stretching
 - b. Image normalization
 - c. Gamma correction with $\gamma < 1$
 - d. Histogram equalization
 - e. Median filtering

4. Consider the following image with alternating white, grey, and black stripes, each exactly 1 pixel wide. You apply a 3×3 median filter to this image. Which image will be the resulting image if you do not apply any border padding?



5. Consider line detection in a noisy edge map with spurious edge pixels. Why is the Hough Transform more suitable over least-squares line fitting that minimizes the average squared distance?
 - a. Least-squares line fitting cannot represent vertical lines because it uses a $y = mx + b$ parameterization, whereas the Hough Transform uses a normal form that can represent all orientations.
 - b. Least-squares line fitting is computationally more expensive than the Hough Transform for large images.
 - c. Least-squares line fitting minimizes a global squared error and is sensitive to outliers, whereas the Hough Transform accumulates local votes in parameter space, making it more robust to scattered noise.
 - d. Least-squares line fitting requires the number of lines to be known in advance, while the Hough Transform does not.
 - e. Least-squares assumes Gaussian noise on the dependent variable y ; this assumption is violated by binary edge maps.

6. Recall the normal parameterization of a line, where $\rho = x \cos \theta + y \sin \theta$. Which of the following parameter pairs $(\rho_1, \theta_1), (\rho_2, \theta_2)$ represents an image with two straight lines perpendicular to each other?
 - a. $\theta_2 = \theta_1$
 - b. $\theta_2 = \theta_1 + \frac{\pi}{2}$
 - c. $\theta_2 = \theta_1 + \pi, \rho_2 = -\rho_1$
 - d. $\theta_2 = \theta_1 + \frac{\pi}{4}, \rho_2 = \rho_1$
 - e. $\theta_2 = \theta_1 + \pi, \rho_2 \neq \rho_1$

7. In the standard circle Hough Transform, where the radius is unknown, the vote accumulator is 3D: (a, b, r) . If the gradient orientation ϕ at an edge point (x_i, y_i) is used to constrain the possible circle centres, what geometric shape would the point's votes form in the 3D accumulator space?
 - a. One 3D cone
 - b. One 2D circle
 - c. Two 2D circles that intersect each other
 - d. Two discrete points
 - e. 2 straight lines projecting through the 3D space

8. You apply the standard Generalized Hough Transform for an irregular object. During offline modelling, you construct a table that stores displacement vectors from edge points to a chosen reference point, indexed by the gradient orientation (this table is commonly called the R-table). Which of the following modified object-model combinations would still result in the correct detection?
 - a. A set of points and a set of lines
 - b. A set of points and a set of circles
 - c. A set of points and a set of ellipses
 - d. A set of points and a set of rectangles
 - e. A set of points and a set of polygons

- a. The object is shifted by $(\Delta x, \Delta y)$; before detection, all stored displacement vectors in the R-table are also offset by $(\Delta x, \Delta y)$.
- b. The object in the scene is uniformly scaled by a factor of 2; before detection, all stored displacement vectors in the R-table are multiplied by 2.
- c. The object in the scene is rotated by 30° clockwise around the reference point. Before detection, the gradient orientations in the R-table are adjusted by subtracting 30° .
- d. The object is mirrored horizontally. Before detection, the displacement vectors are also mirrored horizontally.
- e. The object is rotated by 30° clockwise around the reference point. Before detection, the stored displacement vectors in the R-table are rotated by 30° clockwise.

9. In lectures, the mean-shift algorithm was illustrated using a radially symmetric window with a single bandwidth parameter h , so the same window extent was applied uniformly across all feature dimensions. In practice, the formulation is extended to multiple bandwidth parameters, to allow for different scaling of the feature dimensions.

You apply mean-shift clustering in a joint spatial-colour feature space using bandwidths h_s (spatial) and h_c (colour). The algorithm converges, but the resulting segmentation produces hundreds of small, spatially contiguous regions that differ only slightly in colour. Which adjustment is most appropriate to reduce the number of resulting segments?

- a. Increase h_s ; decrease h_c .
- b. Decrease h_s ; increase h_c .
- c. Keep h_s fixed; increase h_c .
- d. Increase h_s fixed; keep h_c fixed.
- e. Decrease h_s ; decrease h_c .

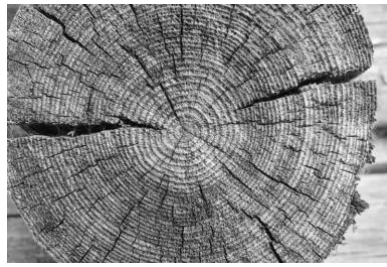
10. A student applies standard K-means clustering in a 5D feature space of (R, G, B, x, y) to segment an image into 100 regions. The result has irregular boundaries and the algorithm converges slowly. They switch to SLIC superpixels instead. What key limitation of standard K-means does SLIC address?

- a. Standard K-means treats the spatial and colour dimensions uniformly; SLIC modifies the distance metric to balance spatial and colour components differently.
- b. Standard K-means evaluates distances from every pixel to every cluster centre globally; SLIC restricts distance computations to a local $2s \times 2s$ region around each centre.
- c. Standard K-means uses Euclidean distance; SLIC replaces it with a perceptually uniform metric in alternate colour spaces.
- d. Standard K-means optimizes a non-convex objective; SLIC reformulates the problem as a convex optimization.
- e. Standard K-means updates cluster centres using the mean; SLIC updates them using the median to improve robustness.

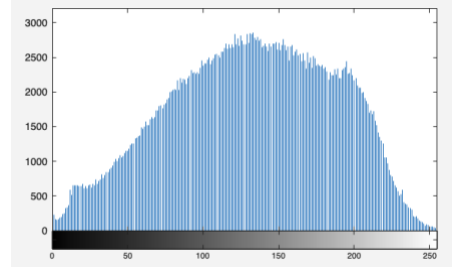
System Design (6 points)



(a) coloured image of tree cross-section



(b) greyscale image of tree cross-section



(c) Intensity histogram of the greyscale image

Consider the image of a cross-section of a tree in (a). It shows growth “rings”, or concentric bands, where each band corresponds to one year of growth. Your task is to count the number of rings (to estimate the tree’s age) and estimate the relative width of the bands (to infer climate conditions for each year).

Begin by converting the image into greyscale. Then, enhance the image to make the rings more prominent. Finally, design a strategy to estimate the number of rings and width of each ring.

Q1: Greyscale conversion (1 point)

If the RGB image has pixel intensities $\{R, G, B\} \in [0,1]$ and the greyscale intensity I is obtained by the linear combination $I = w_R R + w_G G + w_B B$, where $\{w_R, w_G, w_B\}$ are scalar weights. Which of the following are feasible values for $\{w_R, w_G, w_B\}$? Select all that apply.

$\{0.33, 0.33, 0.33\}$, $\{0.5, 0.5, 0.5\}$, $\{0.2, 0.7, 0.1\}$, $\{-0.2, 1.0, 0.2\}$, $\{0.4, 0.4, 0.2\}$, $\{0.1, 0.2, 0.3\}$

Q2: Ring enhancement (1 point)

You want to process the greyscale image so that the darker bands in the rings are more prominent. The image and its corresponding histogram is shown in (b) and (c) respectively. Which of the following operations are suitable? Select all that apply.

Histogram equalization.

Histogram scaling.

Histogram normalization.

Intensity scaling, with $a > 0$.

Gamma correction, with $\gamma < 1$.

Box filtering.

Median filtering with a 3×3 kernel.

Normalized cross-correlation with a circular template.

Q3: Ring Counting and Width Estimation (4 points)

Assume that you start with an image in which the rings are already sufficiently enhanced from pre-processing. Detail a method for estimating the number of rings and the width of each ring; note the rings are not perfectly circular. State key assumptions for your solution and outline potential weaknesses.

